Programming in HTML5 with JavaScript and CSS3 Chapter 14: Making your HTML location-aware

Q1. What are the underlying location information sources of Geolocation?

A. locations inferred from network
B. radio frequency identification (RFID)
C. Wi-Fi and Bluetooth MAC addresses
D. All of the above

Answer: D
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Underlying location information sources such as the Global Positioning System (GPS) and locations inferred from network signals such as IP address, radio frequency identification (RFID), Wi-Fi and Bluetooth MAC addresses, and cellular IDs and from user input]

Q2. Which one of the following provides a way to integrate location services into a web page?

A. The Web storage API
B. The Geolocation API
C. The AppCache API
D. The Media Query

API Answer: B
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Q3. Using the Geolocation API, you can get _____.

A. the current position of the server from which the application is serving the web page to the user device
B. the current position of the user or the device in which the application is running
C. the current position of the nearest GPS device available to the user device
D. the current position of the network server that provides user or the device internet

gateway Answer: B
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Q4. You are creating a web application that will help users finding nearby points of interest or checking into their favorite social community sites. Which feature you will use?

A. The Web storage API
B. The Geolocation API
C. The AppCache API
D. The Media

features Answer: B
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Q5. The Geolocation API is designed to support _____?

A. one-shot position requests only.
B. repeated position updates only.
C. both one-shot position requests and repeated position updates.

D. None of
above Answer: C
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Q6. You get a reference to the Geolocation API using
A. geolocation
B. window.geolocation
C. navigator.geolocation
D. window[‘geolocation’]
Answer: C
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Q7. The Geolocation object is accessible by using the _____ global variable?
A. window.geolocation
B. navigator.geolocation
C. browser.geolocation
D. location.geolocation
Answer: B
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Q8. You want to determine whether the user’s browser provides support Geolocation API. Which of the following code does not accomplish it properly?
A. if('geolocation' in navigator)
{ //code
}
B. if(navigator.geolocation) {
//Code
}
C. if(navigator['geolocation']) {
//Code
}
D. if(geolocation)
{ //Code
}
Answer: D
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Property can be accessed using key in JavaScript
window.location is equivalent to window[‘location’]
Property can be enumerated using foreach in JavaScript
foreach (var prop in window) {//work with prop}
So ‘geolocation’ in navigator will return true if supported otherwise false (null which evaluate to false)]

Q9. Which one of the following is not a valid call to getCurrentPostion method of the Geolocation object?
A. getCurrentPosition()
B. getCurrentPosition(positionCallback)
C. getCurrentPosition(position Callback, positionErrorCallback)
D. getCurrentPosition(positionCallback, positionErrorCallback, positionOptions)

Answer: A
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getCurrentPosition(positionCallback, [positionErrorCallback], [positionOptions])
First parameter is mandatory, other two are optional
position Callback - is called after the current position is determined. positionErrorCallback - is called if any errors occur when trying to get the current position
position Options - lets you set some special options that control how the getCurrentPosition method behaves.]

Q10. Which of the following is not the methods of the Geolocation object?

A. getCurrentPosition()
B. watchPosition()
C. clearWatch()
D. LatLng()

Answer: D
[Page 540
LatLang() is the method of Google map API, actually it is a object constructor]

Q11. What is not a parameter of the Geolocation object's getCurrentPosition method?

A. A success callback
B. An error callback
C. options array
D. timestamp

Answer: D [Page
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Q12. Which method continuously monitors your current location from the Geolocation object?

A. watchPosition()
B. watchLocation()
C. getCurrentPosition()
D. getCurrentLocation()

Answer: A
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Q13. What is the type of the parameter that the success callback of the getCurrentPostion or watchPostion method accepts?

A. Geoposition
B. Coordinate
C. Latitude
D. Longitude

Answer: A
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Q14. Which property of the Geopostion object allows to access information about current position like longitude, latitude etc.?

A. position
B. longitude
C. latitude
D. coords

Answer: D
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Q15. Which of the following are the members of Geoposition object?

A. coords
B. timestamp
C. longitude
D. latitude

Answer: A, B
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Q16. What does the coords property of the Geoposition return?

A. a Coordinates object
B. a LatLng object
C. a Map object
D. a JSON

object Answer: A
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Q17. Which one is not a member of the Coordinate object?

A. latitude
B. longitude
C. altitude
D. enableAccuracy

Answer: D
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latitude - Gets the latitude in decimal degrees
longitude - Gets the longitude in decimal
degrees altitude - Gets the height in meters
accuracy - Gets the accuracy of the coordinates in meters altitudeAccuracy Gets the accuracy of the altitude in meters heading Gets the direction of travel in degrees
speed - Gets the speed of travel in meters/second]

Q18. Which property of the Geoposition of the direction of travel?

A. direction
B. travelDirection
C. heading
D. travel

Answer: C
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Q19. How do you control how the getCurrentPosition method behaves?

A. By setting a PositionOptions object to the Geolocation object's option property
B. By passing a PositionOptions object as the third parameter while calling the getCurrentPosition method
C. By setting a PositionOptions object to the options property of the Geoposition before calling the getCurrentPosition method
D. By passing a PositionOptions object as a parameter to the success callback for the getCurrentPosition method

Answer: B
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Q20. Which property of the PositionOptions object is used to indicate to use a cached result if available?

A. timeout
B. enableHighAccuracy
C. maximumAge
D. cache

Answer: C
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enableHighAccuracy This causes the method to be more resource intensive if set to true. The default is false. If true, the getCurrentPosition method tries to get as close as it can to the actual location.
timeout This specifies a timeout period for how long the getCurrentPosition method can take to complete. This number is measured in milliseconds and defaults to zero. A value of zero represents infinite.
maximumAge If this is set, the API is being told to use a cached result if available, rather than make a new call to get the current position. The default is zero, so a new call is always be made. If maximumAge is set to a value and the cache isn't older than the allowable age, the cached copy is used. This value is measured in milliseconds.]

Q21. What is the default value of the maximumAge property of PositionOptions object?

A. -1
B. 0
C. 1000
D. 5000

Answer: B
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Q22. Which one always get the updated current position?

A. navigator.geolocation.getCurrentPosition((pos) =>{} , (err) =>{});
B. navigator.geolocation.getCurrentPosition((pos) =>{} , (err) =>{}, {timeout:0, maximumAge:1000});
C. navigator.geolocation.getCurrentPosition((pos) =>{} , (err) =>{},{maximumAge:5});
D. navigator.geolocation.getCurrentPosition((pos) =>{} , (err) =>{},{timeout:1000,maximumAge:5000});

Answer: A
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Q23. Which one correctly get the new position if cache is older than 5 seconds?

A. navigator.geolocation.getCurrentPosition((pos) =>{} , (err) =>{});
B. navigator.geolocation.getCurrentPosition((pos) =>{} , (err) =>{}, 5000);
C. navigator.geolocation.getCurrentPosition((pos) =>{} , (err) =>{},{maximumAge:5});
D. navigator.geolocation.getCurrentPosition((pos) =>{} , (err) =>{},{maximumAge:5000}); Answer: D
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Q24. The default value of the maximumAge of the PositionOptions object is set to 0. What does it indicate?

A. It means that a cached location is used if available.
B. It means that a cached location is not used.
C. It means that it will prompt user to decide whether a cached location is to be used or not
D. None of the above

Answer: B
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Q25. Which of the following calls the getCurrentPosition method of the Geolocation object?

A.
```
navigator.geolocation.getCurrentPosition(function (position) {
        //  Code to access co-ordinates
  },
  function (err) {
    //Error handling
  },
  {enableHightAccuracy: true, timeout: 3000,
        maximumAge:20000 });
```
B.
```
navigator.geolocation.getCurrentPosition({enableHigtAccuracy: true, timeout: 3000,
maximumAge: 20000 },
function (position) {
  //Code to access co-ordinates
}, function (err) {
  //Error handling
});
```
C.
```
navigator.geolocation.getCurrentPosition(function (position) {
  // Code to access co-ordinates
},
{enableHighAccuracy: true, timeout: 3000,
maximumAge: 20000 });
```
D.
```
navigator.geolocation.getCurrentPosition(function () {
  //  Code to access co-ordinates ;
}, function (err) {
  //Error handling
},
{enableHighAccuracy: true, timeout: 3000,
maximumAge: 20000 });
```

Answer: A
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Q26. You are working with Geolocation object. You want to know the user's current position. You have the following code: (line numbers are illustrative purpose only)

```
Line 1:     if(navigator['geolocation']) {
Line 2:             navigator.geolocation.getCurrentPosition(function (position) {
Line 3:
Line 4:
Line 5:                     //code to work with latitude and longitude
Line 6:             }, function (err) {
Line 7:                     //Error handling code
```

```
Line 8:                         });
Line 9:         }
```

A. var lat = position.latitude;
   var lng = position.longitude;
B. var lat = position.coords.latitude;
   var lng = position.coords.longitude;
C. var lat = position.LatLng.latitude;
   var lng = position. LatLng.longitude
D. var latlng = new LatLng(position.coords);
   var lat = latlng.latitude;
   var lng = latlng.longitude;

Answer: B
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Q27. Which method of the Geolocation object is used to retrieve continuous position updates?
A. getCurrentPosition
B. watchPosition
C. clearWatch
D. watch

Answer: B
[Page 546]

Q28. Which of the following is or are true?
A. getCurrentPosition() of the Geolocation object calls one-time the success callback with the current position
B. getCurrentPosition() of the Geolocation object calls continuously calls the success callback with the current position
C. watchPosition() of the Geolocation object calls one-time the success callback with the current
D. watchPosition() of the Geolocation object calls continuously calls the success callback with the current position

Answer: A, D
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Q29. Which of the following formulas can you use to calculate the distance between two samples?
A. haversine
B. Pythagorean theorem
C. quadratic
D. hyperbolic

Answer: A
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Q30. What does the watchPosition method return?
A. A Geoposition object
B. A Coordinate object
C. A PositionOptions object
D. An id which is used in clearWatch method to stop monitoring user’s location

Answer: A

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Q31. Which method monitors your location?

A. getCurrentPosition
B. watchPosition
C. clearWatch
D. watch

Answer: B
[Page 549]

Q32. How do create a Map object using Google Map API?

A. By using google.Map class
B. By using google.maps.Map class
C. By using google.maps.LatLng
D. By using Map class

Answer: B
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Q33. You want to show Google map pointing user position in your application. You have the following code

```
1. navigator.getCurrentPosition ((position)=>{
2.
3. });
```

A. var map = new google.maps.Map(document.getElementById('map'),position);
B. var map = new google.maps.Map(document.getElementById('map'), position.latitude,position.longitude);
C. var map = new google.maps.Map(document.getElementById('map'), {lat: position.latitude, lng: position.longitude });
D. var map = new google.maps.Map(document.getElementById('map'), {
center: {lat: position.latitude, lng: position.longitude}});

Answer: D
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Google Map constructor
google.maps.Map(mapcontainer, [mapOption])
MapOption center property is LatLng set objects
LatLng.lat - latitude
LatLng.lng - longitude]